Learning for Graph-Based Sensorless Freehand 3D Ultrasound

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Abstract. Sensorless freehand 3D ultrasound (US) uses speckle decorrelation to estimate small rigid motions between pairs of 2D images. Trajectory estimation combines these motion estimates to obtain the position each image relative to the first. This is prone to the accumulation of measurement bias. Whereas previous work concentrated on correcting biases at the source, this paper proposes to reduce error accumulation by carefully choosing the set of measurements used for trajectory estimation. An undirected graph is created with frames as vertices and motion measurements as edges. Using constrained shortest paths in the graph, random trajectories are generated and averaged to obtain trajectory estimate and uncertainty. To improve accuracy, a Gaussian process regressor is trained on tracked US sequences to predict systematic motion measurement error, which is then used to weigh the edges of the graph. Results on speckle phantom imagery show significantly improved trajectory estimates in comparison with the state-of-the-art, promising accurate volumetric reconstruction.

Keywords: Freehand 3D ultrasound · Speckle decorrelation · Sensorless reconstruction · Measurement selection · Gaussian process regression

1 Introduction

Compared to traditional 2D ultrasound (US), 3D US allows a better evaluation of the anatomical structures. A 3D probe can be used to obtain US volumes, but it is expensive and offers a limited field of view. Using freehand 3D US with a conventional 2D probe, the sonographer sweeps the area of interest to acquire a sequence of 2D frames. Volume reconstruction of this sequence requires the six degree of freedom (6DoF) motions of all the frames relative to the first one. During the acquisition, the probe can be tracked by a position sensor, but this imposes extra cost, is cumbersome and requires calibration. Instead, using the information of the images themselves to perform a sensorless registration of the frames [2–4,8,17] allows easy and inexpensive transducer tracking.

This is possible because speckle patterns are correlated out-of-plane in nearby images. Under Rayleigh scattering conditions, this phenomenon is entirely predictable from transducer characteristics and can be used to measure elevational

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displacement [3]. This approach can be adapted to work outside Rayleigh scattering conditions, allowing measurements in imagery of real tissues [2,6,12]. In-plane displacement can be estimated from the position of the correlation peak. By dividing the image into patches, each with a local 3D displacement estimate, the global 6DoF rigid motion can be estimated by a least squares approach [17].

Freehand motion does not produce pure elevational translations. Unfortunately, other types of motion (particularly rotations), also affect speckle correlation, leading to biased motion estimates [13]. Moreover, elevational translation measurements are limited by speckle correlation range, so we need to combine multiple measurements to estimate each frame's position relative to the first [8]. This results in the accumulation of bias and strongly affects the accuracy of the trajectory estimate. This is the challenge addressed in this paper.

A few papers proposed to correct measurement biases at the source. Correction of the elevational decorrelation curve was proposed to increase accuracy under probe rotation [1,9]. It was also suggested that because the measurement biases are systematic, corrections can be learned from tracked synthetic image sequences to improve the accuracy of sensorless motion estimation [4]. However, we found this approach difficult to generalize to real imagery.

Our approach to the problem is to reduce error accumulation by carefully selecting the measurements used for trajectory estimation. For this purpose, previous work has proposed using maximally spaced images (thereby reducing the number of error accumulation steps) [8], or fusing multiple redundant elevational motion estimates along with measurements of their uncertainty [10]. A hypothesis testing approach was also proposed to eliminate biased measurements arising from inaccurate tissue models used outside Rayleigh scattering conditions [11]. None of these works considered how to choose the *best* motion estimates for trajectory estimate.

We hypothesize that wisely choosing the motions estimated by speckle decorrelation *and* minimizing the number of measurements involved improves probe trajectory estimation. To this end, a graph-based trajectory estimate method, along with a learning-based technique for characterizing measurement error, are proposed in Sect. 2. Experimental results illustrating the benefits of this approach are given in Sect. 3. Section 4 summarizes our contribution.

2 Method

A 2D probe is used to acquire a sequence of N frames $\{f_0, f_1, ...f_N\}$ (Fig. 1a). The estimated position of frame N is denoted $M_N = m_{0N}$, where $m_{ij} \in SE(3),\ i, j \in \{0, ..., N\}$, is the rigid motion between f_i and f_j (Fig. 1b). To simplify experimental validation, we assume that probe motion is monotonic, that there are no intersection between frames, and that we are under Rayleigh scattering conditions. These assumptions could easily be dropped by using methods proposed in the literature [2,6,8,10,12] without invalidating the methods proposed here.

To reduce the error in $M_{k=1,...,N}$, graph-based trajectory estimation from the m_{ij} is proposed (Fig. 1). We first estimate the motion m_{ij} between all pairs of

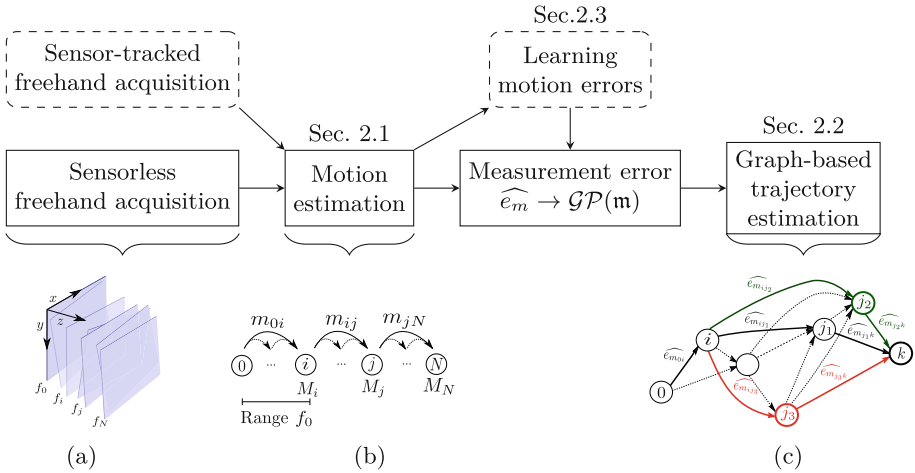


Fig. 1. Block diagram of the method including training (dashed) and testing (solid). (a) Freehand acquisition. (b) Sensorless motion estimation. (c) Reconstruction of frame f_k for $n = 3$ constrained shortest paths through j_1, j_2, j_3.

frames $\{f_i, f_j\}$ using speckle decorrelation (Sect. 2.1). A directed graph is then constructed with frames as vertices and the edges representing the motion estimates between pairs of frames. Using a Gaussian process relating measurement error to the parameters of m_{ij} in a tracked training sequence, it is possible to estimate the error of our sensorless measurements (Sect. 2.3), which we use to weigh the edges of the graph. To retrieve the position M_k of each frame f_k, we generate n random trajectories using a constrained shortest path algorithm and estimate their mean and variance (Sect. 2.2).

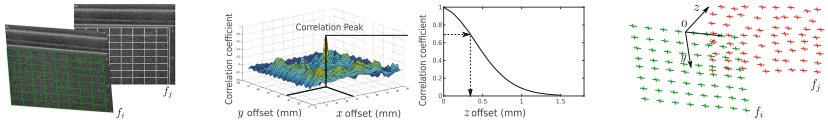
2.1 Rigid Motion Estimation

Estimating motion between pairs of frames requires calibrating a probe-specific elevational speckle decorrelation model [17]. Thus, a speckle phantom is scanned at fixed regular elevational intervals. The images are divided into p non-overlapping patches and the normalized cross-correlation between pairs of corresponding patches is measured. Measurements corresponding to identical elevational displacements are averaged, yielding p local empirical speckle decorrelation curves.

For a new pair of frames $\{f_i, f_j\}$, the in-plane motion of each patch is estimated by maximizing correlation and its out-of-plane motion is read off the local speckle decorrelation curve at peak correlation. The rigid motion $m_{i,j}$ is found from these p local 3D translation measurements (Fig. 2) using least squares [18].

2.2 Graph-Based Trajectory Estimation

Given the motion measurements m_{ij} of a sequence of frames, we want to estimate M_k. One possibility is to compose consecutive pairwise transformations:



(a) (b) (c)

Fig. 2. (a) US image divided in p patches. (b) 3D translation estimate of one patch using speckle decorrelation. (c) Estimation of $m_{i,j}$ by Procrustes alignment.

$$M_k = m_{0,k} = m_{0,1} \circ m_{1,2} \circ ... \circ m_{k-1,k}. \tag{1}$$

This is error prone because (1) empirical decorrelation curves are inaccurate for very short distances and (2) it accumulates biases over many measurements.

The set of motion measurements can be described by a directed unweighted graph, whose vertices represent the frames and whose edges are available motion estimates between pairs of frames. Then, the estimated position of f_k minimizing the accumulation of biases is Dijkstra's shortest path from vertex 0 to vertex k.

Averaging Multiple Solutions. Assuming that motion measurements are subject to random noise as well as bias, we further reduce error by averaging n positions close to the optimal one. Trajectory i is obtained by randomly selecting a vertex $0 < j_i < k$ and finding the shortest path from vertex 0 to vertex k that uses j_i. To average the n positions of f_k, we use Govindu et al.'s intrinsic averaging [7] with the shortest path to f_k as an initial guess. We also compute the standard deviation for the six rigid motion parameters, which is related to the error of the estimated position M_k.

2.3 Learning to Characterize Measurement Error

We can improve measurement selection by weighting the graph edges by measurement error. A typical error measure for sensorless freehand 3D US trajectory estimation is the mean target registration error (mTRE) [5] with respect to position sensor data. In our context, such data are not available on-line. However, there is a predictable relationship between the motion parameters and the measurement error [4]. Thus, we hypothesize that there exists a function g such that $\text{mTRE} = g(m)$, where m is a sensorless motion measurement. Knowing input-output pairs $\{m_{i,j}, \text{mTRE}_{m_{i,j}}\}$ from an accurately tracked sequence, g could, in principle, be learned. However, position sensors are not sufficiently accurate to measure the small motion between two consecutive frames of a typical freehand 3D US acquisition ($\sim 5 \times 10^{-2}$ mm).

Error Measure. The first task is to define an error measure e for a sensorless motion measurement m, knowing the position sensor output. The idea is to create a window bounded by two frames f_l and f_k such that the displacement $d_{\kappa\lambda}$ between the two frames leads to negligible sensor error, i.e., $d_{\kappa\lambda} > 2$ mm. We compute n trajectory estimates $\{\widehat{T_{\kappa\lambda_1}} ... \widehat{T_{\kappa\lambda_n}}\}$ from vertex κ to vertex λ using the constrained random positions of Sect. 2.2 and compute their errors

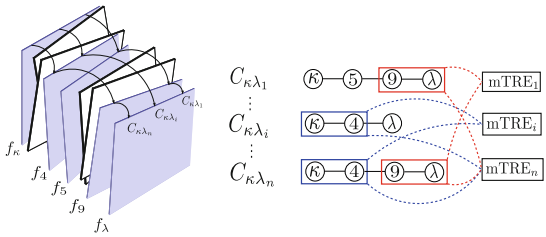


Fig. 3. Assigning mTREs to motions $m_{\kappa,4}$ (blue) and $m_{9,\lambda}$ (red) for window $[\kappa, \lambda]$. (Color figure online)

$\{\mathrm{mTRE}_1 ... \mathrm{mTRE}_n\}$ with respect to the sensor measurement T_{lk}. The computation of one transformation $\widehat{T_{\kappa\lambda_i}} = m_{\kappa,x} \circ ... \circ m_{y,\lambda}$ biased by mTRE_i involves many pairwise motion measurements. The process is repeated with different windows and the *error* $e_{m_{i,j}}$ of each sensorless motion measurement m_{ij} is defined as the average mTRE of all the window-wise position estimates it was involved in (Fig. 3).

Error Prediction Model. Finding a function $e_m = g(m)$ that relates the estimated motion $m_{i,j}$ to its error $e_{m_{i,j}}$ is a non-linear regression problem which we solve using Gaussian processes ($\mathcal{GP}$) [14]. $\mathcal{GP}$ regression is a Bayesian approach which assumes that $g(m)$ behaves according to $p(g|m_1, ..., m_n) = \mathcal{N}(\mathcal{M}, \mathcal{K})$, where $\mathcal{M}$ is the mean vector and $\mathcal{K}$ is the covariance function of the motions m. We used a pure quadratic mean function and a squared exponential covariance function to capture the smoothness of e_m. Error on inputs m, outputs e_m and lengthscale l were trained using quasi-newton optimization with the MATLAB Statistics and Machine Learning Toolbox. To have independency between the parameters of the rotation, we mapped the motion from Lie group $SE(3)$ to Lie Algebra $\mathfrak{se}(3)$ with the log function $\mathfrak{m} = \log(\mathrm{m})$ [7].

3 Experiments and Results

3.1 Data Acquisition

Images of a speckle phantom were acquired using an ATL HDI5000 US scanner with a linear 4–7 MHz transducer at a depth of 3 cm. The phantom was first scanned elevationally over 50 mm at regular 0.05 mm intervals by mounting the transducer on a robot arm following a purely translational trajectory, controlled by a step motor with a precision of 5 µm. The images were divided into 8×8 patches of size 55×32 pixels (Fig. 2a) and the speckle decorrelation curve corresponding to each patch was computed as described in Sect. 2.1.

Nine freehand sequences tracked with a Micron Tracker optical sensor were then acquired with the transducer moving mainly elevationally, with total displacements up to 30 mm. The optically tracked transducer was calibrated temporally and spatially using our adaptation of Rousseau et al.'s plane phantom methods [15,16]. We found experimentally that this setup produces volume measurements within 97 % accuracy.

3.2 Evaluation of Probe Trajectory Estimation

We evaluated five approaches for probe trajectory estimation: (1) the nearest neighbor (NN) approach (Eq. 1); (2) Housden *et al.*'s approach [8] which generates a coarse sequence (CSq) with blocks of three nearly equally spaced frames; (3) the farthest neighbor approach (FN), which takes the first and third frames of each block of CSq, and is better suited to monotonic sequences; (4) the average unweighted graph-based trajectory estimate (GA) of Sect. 2.2; (5) the proposed approach (GMeA) which is GA weighted by measurement error (Sect. 2.3). For GMeA, the $\mathcal{GP}$ was trained using one freehand sequence.

Each approach was tested on the eight other sequences, labeled S1-S8 with $n = 1000$ random positions for GA and GMeA. Table 1 shows the mTRE of the last frame of each sequence (measured at the patch centers) with respect to the optical sensor data, along with the total elevational displacement $|z|$ of the probe. GMeA improves accuracy by 20 % compared to FN. For all methods, error increases with $|z|$. This is expected as longer sequences require more measurements for position estimation and error accumulation cannot be avoided entirely. Some trajectories are clearly estimated less accurately than others (e.g., S2 vs S6), probably due to differences in rotational motion.

Table 1. mTRE of the last frame in S1-S8, as estimated by each method with total elevational displacement $|z|$ for reference and best results in bold.

	S1	S2	S3	S4	S5	S6	S7	S8
total $\|z\|$	20 mm	21 mm	25 mm	27 mm	36 mm	38 mm	52 mm	56 mm
NN	11.041	11.866	15.492	13.897	10.699	12.790	15.321	18.973
CSq	4.395	3.816	5.820	4.683	6.207	10.531	12.655	13.713
FN	2.036	1.028	3.758	2.412	4.758	10.486	10.578	10.999
GA	**1.781**	0.767	3.488	2.055	3.961	10.160	9.790	10.503
GMeA	1.803	**0.731**	**3.039**	**1.995**	**3.735**	**9.345**	**9.154**	**9.999**

To perform meaningful comparisons over the methods and image sequences, the mTREs of all the frames in each sequence (measured at the patch centers) were normalized by the total probe displacement. A Kolmogorov-Smirnov test showed that the results are not normally distributed. Thus, we performed a Friedman test to evaluate the influence of the methods on each sequence, except NN and CSq which are clearly less accurate than the others (Table 1). A Tukey contrast on the ranked means tested the hypothesis that the approaches GA and GMeA have the same accuracy as FN. Because we worked on minimizing drift, the effect of our methods is stronger towards the end of each sequence. A statistically significant difference between GMeA and FN was found when considering the last 35 frames of each sequence ($p < 0.05$), and for the last 10 frames ($p \ll 0.001$). GA was not found significantly different from FN.

Figure 4 shows detailed results for sequence S4. CSq over-estimates rotations and under-estimates elevational translations because there is confusion between

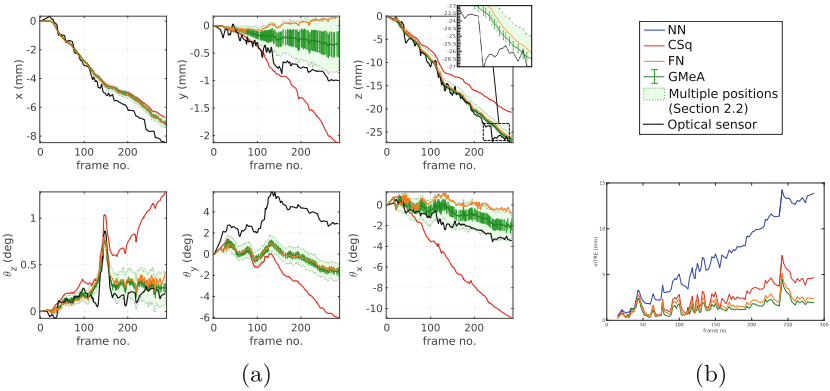


Fig. 4. (a) Freehand trajectories of (CSq, FN and GMeA) for the six degrees of freedom in sequence S4. (b) Error comparison of (NN, CSq, FN, GMeA).

rotation and translation induced decorrelation [9]. The averaging step of GMeA improves the rotational and translational estimates, whereas FN has more difficulty following the optical sensor data. Choosing the best measurements in the graph, as characterized by GMeA, improves the trajectory estimate compared to GA. The standard deviations of the estimated position parameters are a measure of uncertainty for the 3D reconstruction step. They grow with time as there are more plausible motion measurements to choose from. This difficulty is intrinsic to the sensorless approach. Qualitative results shown in Fig. 5 highlight the improved accuracy of volumetric reconstruction with GMeA versus CSq compared to that obtained with the optical sensor.

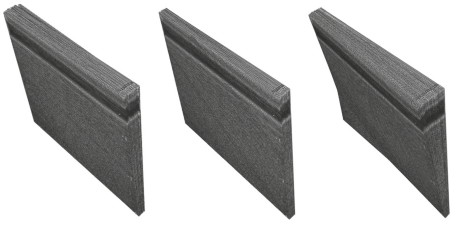


Fig. 5. (Left to right) Volumetric reconstruction of sequence S4 with optical sensor, GMeA and CSq.

4 Conclusion

This paper shows that measurement error can effectively be learned from tracked data and used towards improved sensorless freehand 3D US reconstruction. Our error weighted graph-based method significantly improves over the state-of-the-art. Future work will concentrate on adapting our method to non-Rayleigh scattering conditions.

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